Media, Motion, and the Body

Project 1: Circuits and Interactions

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Design Statement

In this project, I explored how the body can act as an interface through physical computing. I created an interactive hopscotch mat that translates jumping and foot placement into sound using copper tape, a Makey Makey, and p5.js.

I chose hopscotch because it is a familiar, intuitive game that already carries an understanding of movement, rhythm, and balance.

- Rather than asking users to learn a new interface, the design relies on the existing physical knowledge of stepping, hopping, and landing to trigger interaction.
- Each jump closes a physical circuit, using the Makey Makey it sends a signal to the computer that produces immediate audio feedback, reinforcing the connection between action and response.

Through this work, I focused on making the interaction feel playful, legible, and responsive, while revealing how simple materials and gestures can form interactive systems.

Research - Concept 1

Concept – Affordances & Embodied Interaction

Affordances describe the actions an object or space suggests through its form and context.

- James Gibson defines affordances as possibilities for action that are directly perceived, not learned through instruction.
- In embodied interaction, meaning emerges through physical movement, where the body itself becomes the interface.

Application to my project

- Hopscotch already carries a culturally learned affordance: jump, land, balance, move forward.
- I chose a familiar game so users could interact without instructions or explanation.
- The mat's scale, spacing, and layout naturally guide movement through the body.
- Sound feedback confirms successful interaction, reinforcing learning through movement and repetition.

Research - Concept 2

Concept – Time, Response, and Physical Computing Systems

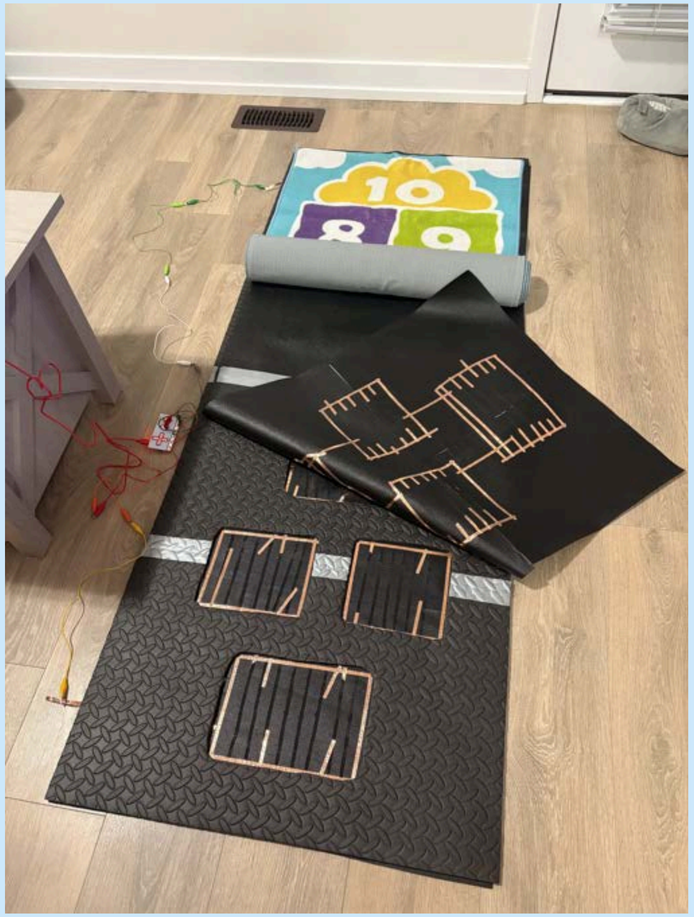
Lectures emphasized interaction as something that unfolds over time, not instant outcomes.

- Physical computing systems respond to movement, pressure, and sequence, rather than single inputs.
- Sound and motion change make invisible systems perceptible to the user.
- Iteration and testing reveal how timing and response shape experience.

Application to my project

- Jumping activates a circuit that produces sound and visual change.
- The timing between foot placement, pressure, and response became part of the interaction.
- Delays, missed triggers, and accidental activations informed design adjustments.
- Iteration focused on tuning sensitivity so response felt intentional and readable.

Final Design and Demo



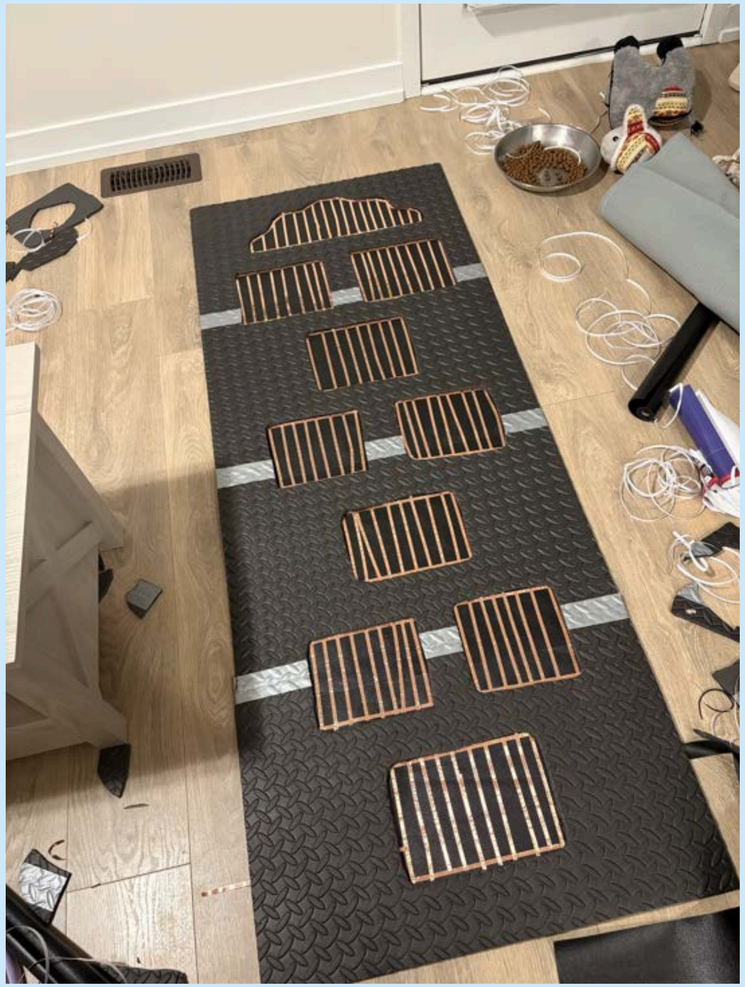
Challenges

Materials & Physical Constraints

- Early prototypes were designed for hand pressure, not full body weight.
- When hopping, too much copper tape caused over-sensitivity and accidental triggering.
- Through testing, I learned that less copper tape required less pressure and gave me more control.
- Iteration involved repeatedly adding and then removing tape to balance sensitivity.

Interaction & Timing

- Sound output depended on foot placement, timing, and landing rhythm.
- Jumping is not a single action but a sequence of movements, which affected reliability.
- Missed triggers helped me refine spacing, pressure points, and interaction pacing.

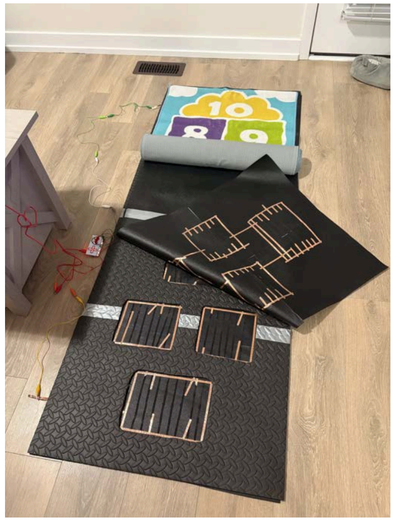


Overcoming Challenges

Most of my learning came from failure!

Key learning:

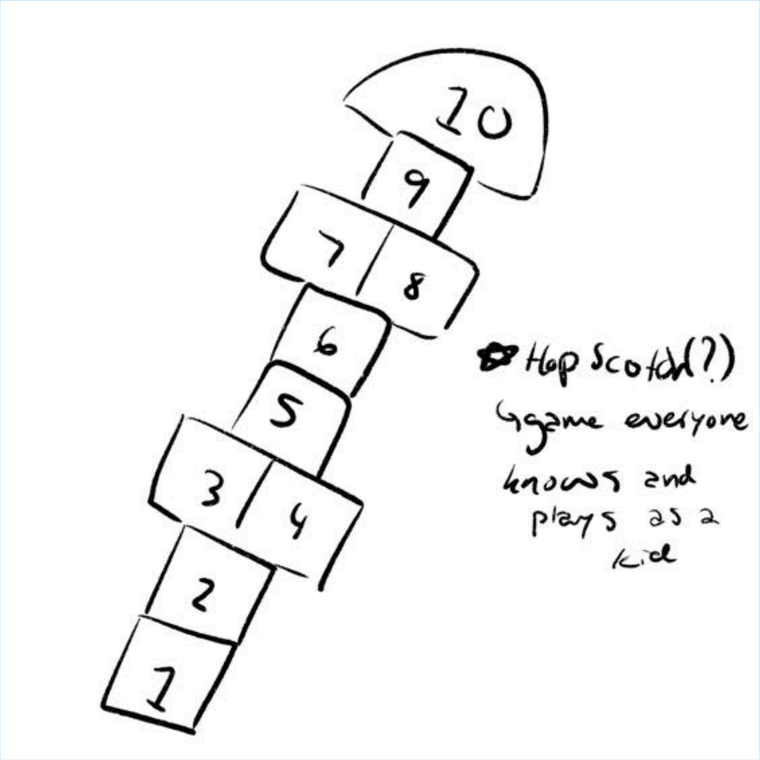
- Iteration revealed that material sensitivity, timing, and restraint were essential to making the interaction legible.



Thinking Process

- I began by exploring how simple physical gestures could become interactive inputs.
- Early ideas focused on circuits responding to touch, pressure, and movement.
- Through formstorming and material tests, I realized I was most interested in full-body interaction, not hand-based input.
- Hopscotch emerged as a strong direction because it already combines movement, rhythm, balance, and sequence.

Rather than inventing a new gesture, I chose to work with a familiar physical language and see how it could be translated into an interactive system.



Reflection

This project changed how I think about interaction design as something that is felt through the body, not just seen on a screen.

I learned that material sensitivity, timing, and restraint are critical when designing for physical input.

Iteration taught me that failure is informative, especially when working with physical computing systems.

Debugging both the circuit and the code helped me understand how small technical details shape user experience.

Overall, this project strengthened my confidence working across physical materials, code, and embodied interaction, and reinforced the importance of testing with real bodies, not assumptions.

Final Takeaway

Designing with the body requires listening to how systems respond to movement, weight, and time.

Work Cited

OpenAI. ChatGPT. Version GPT-5.2, OpenAI, 2026, https://chat.openai.com/.

- Prompted and used for code debugging.